

Calculus A2 Homework

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1 HW Due Mar 9

Problem 1.1. You will get full credit for this problem as long as you submit this HW, no matter how you answer these questions.

1. What is your expected letter grade for this class?
2. How many hours per week do you plan to commit for this class?
3. Name a friend in this class you could discuss math with.
4. Name a friend that has already aced this class so you can ask questions from.
5. Name your favorite AI model as your personal assistant in all matters. (If you can afford it, use a paid version of whatever AI model. It is worth the investment in my opinion.)
6. No late HW will be accepted, unless you have doctor's notes or special permission from the university. Write "YES" for this subproblem to indicate that you understand.

Problem 1.2. You automatically get full credit for this problem.

1. Check out <https://tutorial.math.lamar.edu/ProblemsNS/CalcIII/CalcIII.aspx>. Alternative notes, exercises, and step-by-step solutions!
2. Check out <https://www.math.utoronto.ca/courses/mat237y1/20199/contents.html>. Alternative notes and exercises.
3. Check out <https://www.youtube.com/watch?v=TrcCbWwCBc&list=PLSQ10a2vh4HC5feHa6Rc5c0wbRTx56nF7>. Khan academy videos.
4. Check out <https://www.xiazr20.top/xiazr20/calculus-ii/.html#3>. This is a Chinese content, made by TA of past Tsinghua Chinese Calculus A2 classes by professor Xiaofeng Wang.
5. Check out [https://github.com/PKUanonym/REKCARC-TSC-UHT/tree/master//A\(2\)](https://github.com/PKUanonym/REKCARC-TSC-UHT/tree/master//A(2)). All Tsinghua exams in the past. However, warning. My exams will take a different style than most other professors.
6. Reflect on your own internet searching ability. Can you gather info/resources quickly and effectly about whatever subject, especially with the help of AI?
7. Finding the resources is only half the battle. USING the resources is the other half, and you should have a plan on how to use them. (But if you use multiple sources or info, beware of potential notation/definition differences and such. You should always standardize to our notation/definitions for our classs.)

2 HW Due Mar 16

Problem 2.1. Match the level curves and the graph of two-variable functions, and give a brief reason for each. (One sentence or two is fine.)

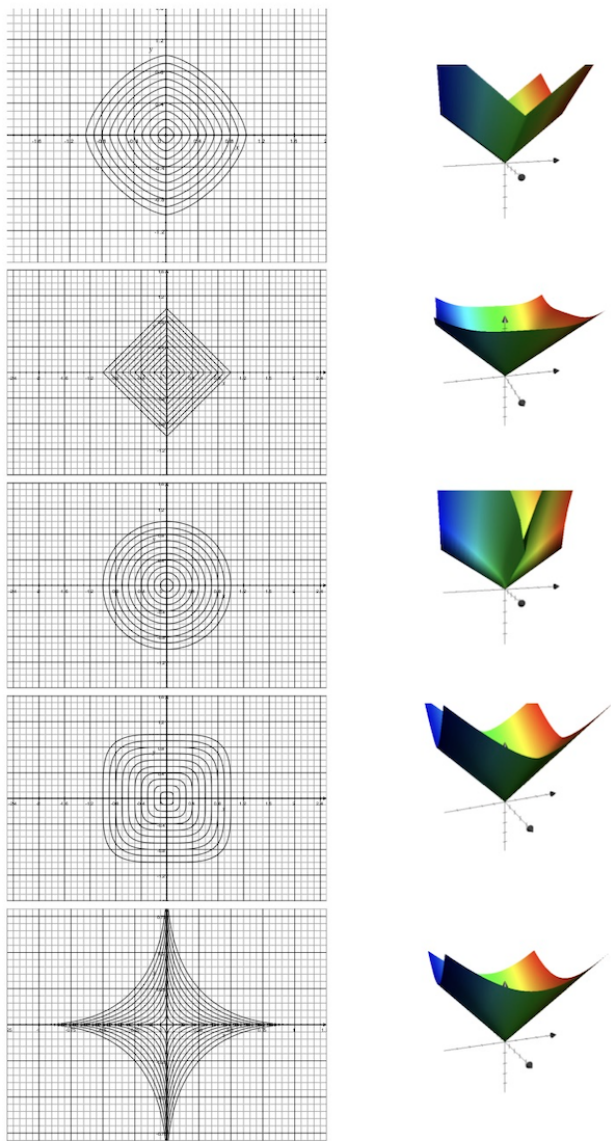
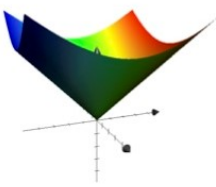
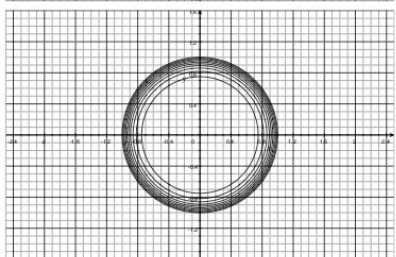
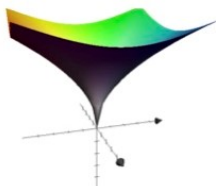
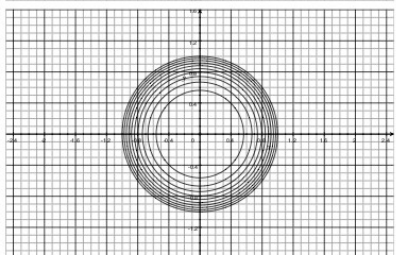
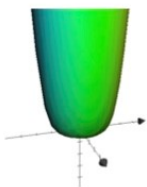
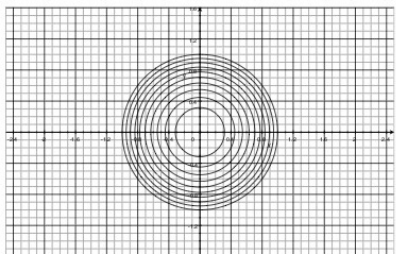
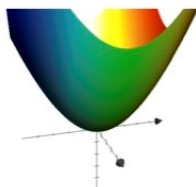
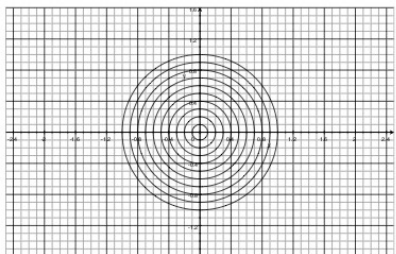
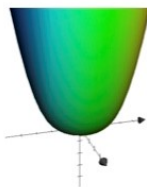
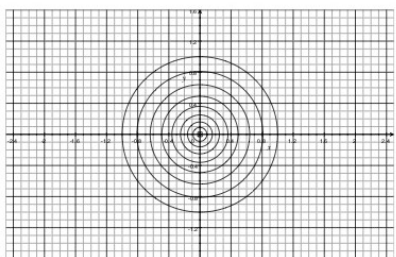
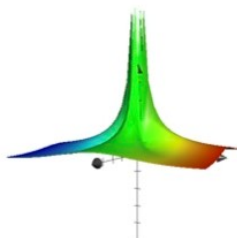
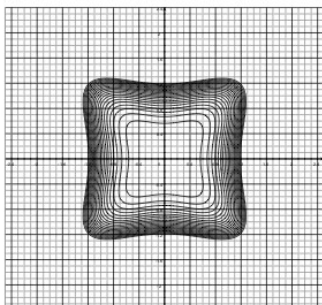
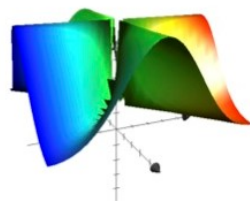
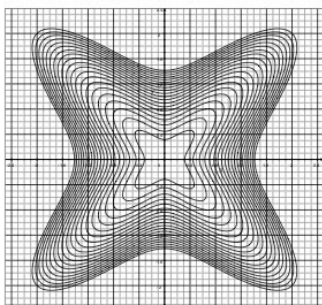
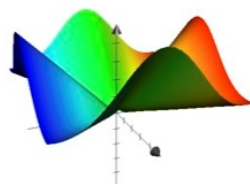
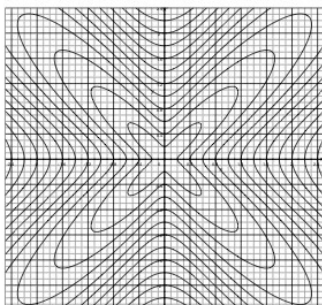
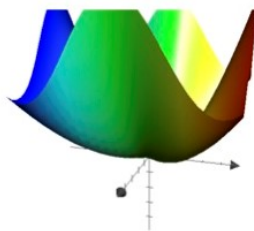
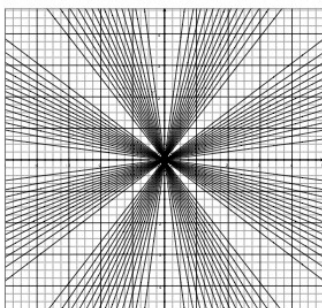
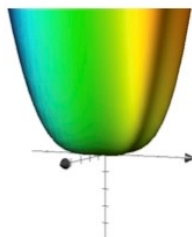
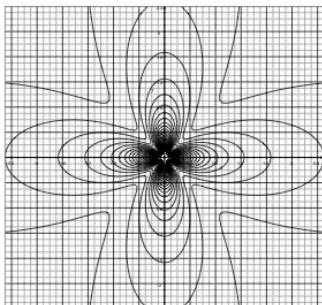


FIGURE 1. match the contour plots on the left with the graphs on the right

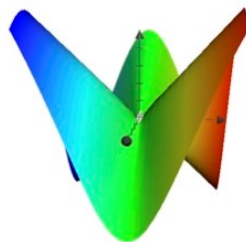
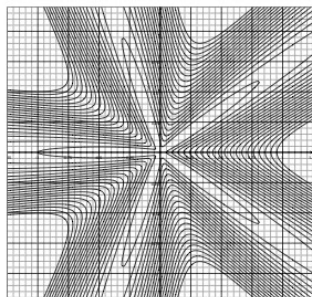
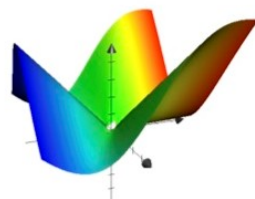
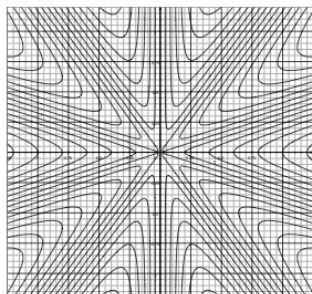
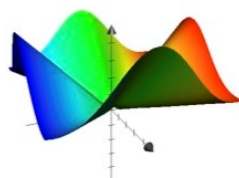
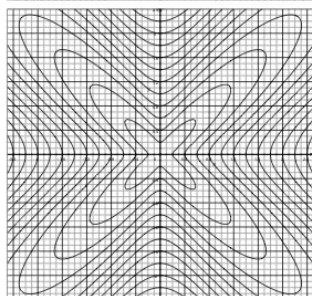
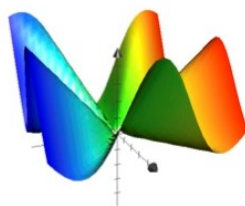
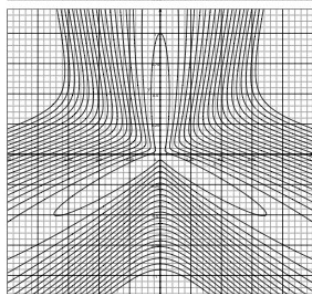
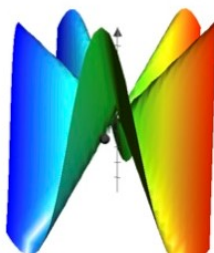
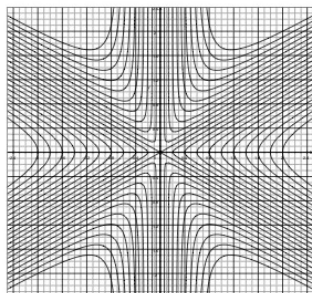
Problem 2.2. Match the level curves and the graph of two-variable functions. Density matters! And give a brief reason for each matching. (One sentence or two is fine.)



Problem 2.3. Match the level curves and the graph of two-variable functions, and give a brief reason for each. (One sentence or two is fine.)



Problem 2.4. Match the level curves and the graph of two-variable functions, and give a brief reason for each. (One sentence or two is fine.) Pay attention to the z -coordinates above and below the xy -plane.

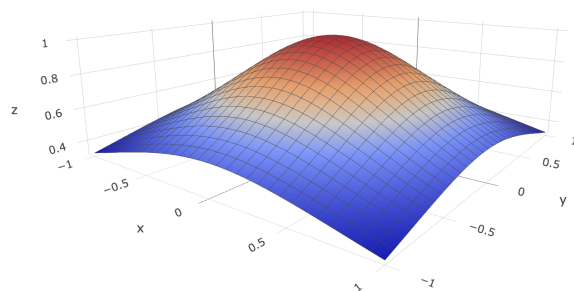


Problem 2.5. Which type of surfaces are these? (E.g., a sphere, an ellipsoid, a paraboloid, a plane, etc..) (The key is to look at the “slices” when you fix z , or maybe when you fix x , or maybe when you fix y .)

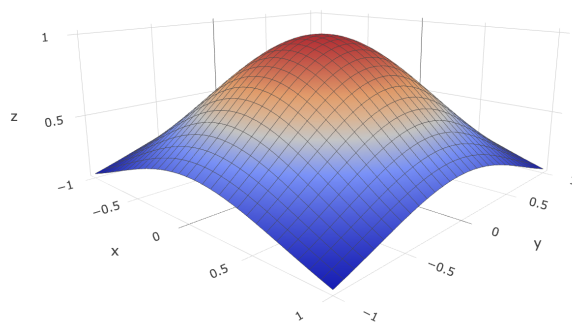
1. $x^2 - 4z - y = 2$.
2. $z^2 + 4x^2 = 1 - 4y^2$.
3. $z + 4x^2 = 1 - 4y^2$.
4. $z - 4x^2 = 1 - 4y^2$.

Problem 2.6. Match the functions and the graphs, and give a brief reason for each. (One sentence or two is fine.)

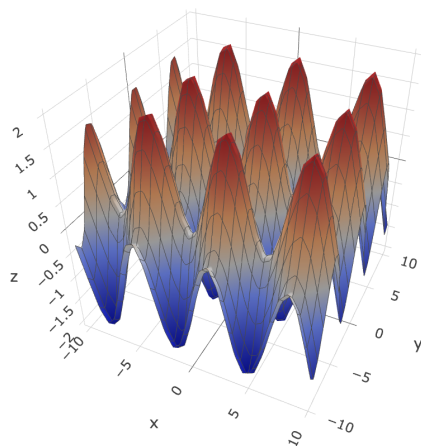
1. $f(x, y) = x^3 - 3xy^2$.
2. $f(x, y) = e^{-x^2 - y^2}$.
3. $f(x, y) = \sin x + \cos y$.
4. $f(x, y) = \frac{1}{x^2 + y^2 + 1}$.



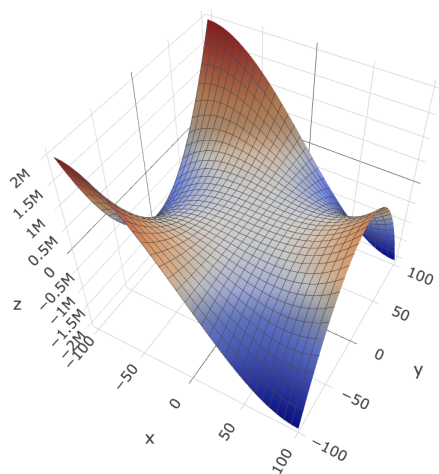
1. A bump.



2. A sharper bump.



3. Egg carton.



4. Monkey Saddle.

Problem 2.7. Write out the equations (in Cartesian coordinates) for the following surfaces in $\mathbb{R}^3$. The expression should only use POLYNOMIALS of x, y, z !

1. A sphere with center $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and radius 2.
2. An infinite cylinder around the line $x = y = z$ and radius 1.
3. A surface with cylindrical equation $z = r^2 \cos(2\theta)$. (Use the double angle formula.) This surface will go “up-down-up-down” as you travel around the z -axis.
4. A surface with cylindrical equation $z = r^3 \cos(3\theta)$. (Use the triple angle formula.) This surface will go “up-down-up-down-up-down” as you travel around the z -axis.
5. A surface with spherical equation $\rho \sin \phi = 2$.
6. A surface with spherical equation $\rho \cos \phi = 2$.
7. A surface with spherical equation $\rho \sin^2 \phi = \cos \phi$.

Problem 2.8. The following parametric expression describes the hypocycloid of a unit circle (b is a constant):

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (1-b)\cos t + b\cos(\frac{1-b}{b}t) \\ (1-b)\sin t - b\sin(\frac{1-b}{b}t) \end{bmatrix}.$$

Please plot this curve for $b = \frac{3}{10}$ with the help of whatever software. Is this a regular parametrization?

(Not part of the problem) For curious students, here is an explanation of why this represents a hypocycloid. Consider a unit circle, with a smaller inner circle of radius b rolling along its interior. The center of this inner circle rotates around a circle of radius $1-b$. Therefore, the trajectory of the center of the inner circle is:

$$\begin{bmatrix} (1-b)\cos t \\ (1-b)\sin t \end{bmatrix}.$$

On the other hand, assume that when the inner circle is at the far right (with its center at $\begin{bmatrix} 1-b \\ 0 \end{bmatrix}$), we mark the rightmost point of the inner circle. As the inner circle begins to roll and reaches the position where the center is $\begin{bmatrix} (1-b)\cos t \\ (1-b)\sin t \end{bmatrix}$, the arc length traveled along the contact interface between the two circles is exactly t .

This means the marked point on the inner circle has also traversed an arc length of t on the inner circle, which corresponds to an angle of $\frac{t}{b}$ within the inner circle. By subtracting this angular displacement from the inner circle's own rotation angle t , we obtain the angular position of the marked point as $-\frac{1-b}{b}t$. Adding the radius vector of the inner circle to its center position yields the final answer:

$$\begin{bmatrix} (1-b)\cos t + b\cos(\frac{1-b}{b}t) \\ (1-b)\sin t - b\sin(\frac{1-b}{b}t) \end{bmatrix}.$$

